SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1710

Name : Preethi R Regno: 192511352

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Preethi R (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Preethi

20 + 20 + 23 + 19 + 14

= 96

Student 24: Preethi R | Register Number: 192511352

Question 1 - Medical Attention Alert

Knowledge Base

1. If a patient has a fever, then the patient needs medical attention.
2. Ravi has a fever.

Goal

Prove that Ravi needs medical attention. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Library Late Fine

Knowledge Base

1. If a book is returned late, then a fine is charged.
2. The book was returned late.

Goal

Prove that a fine is charged. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Flight Delay Notification

Knowledge Base

1. If a flight is delayed, then passengers are notified.
2. Flight 202 is delayed.

Goal

Prove that passengers are notified. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Employee Certification

Knowledge Base

1. If an employee completes the training, then the employee receives a certification.
2. Kevin completed the training.

Goal

Prove that Kevin receives a certification. using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access. using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step and write the final conclusion.

1. Medical Attention Alert

→ propositional logic

Let

F = Ravi has fever

M = Ravi needs medical attention

rule :

$F \rightarrow M$

Fact :

F

Goal : M

→ CNF and negation of Goal

Convert implication :

$F \rightarrow M \equiv \neg F \vee M$

CNF : $\neg F \vee M$

Fact : F

Negated goal :

$\neg M$

→ Resolution

Resolve : $\neg F \vee M$

with F

M

Now resolve : M

$\neg M$

Ravi needs medical attention

2. Library late fine

→ propositional logic

L = Book is returned late

F = Fine is charged

rule:

$L \rightarrow F$

Fact: L

Goal: F

→ CNF

$L \rightarrow F$

$\neg L \vee F$

Fact:

L

Negated goal: $\neg F$

→ Resolution

$\neg L \vee F$

Resolve with: L

Therefore: F

Resolve with $\neg F$

A fine is charged.

3. Flight Delay Notification

→ propositional logic

D = Flight 202 is delayed.

N = Passengers are notified

rule:

$D \rightarrow N$

Fact: D

Goal: N

→ CNF

$D \rightarrow N$

$\neg D \vee N$

Fact: D

Negated goal: $\neg N$

→ Resolution

$\neg D \vee N$

Resolve with: D

Therefore: N

Resolve with: $\neg N$

Passengers are notified.

4. Employee Certification

→ propositional logic

T = Kevin completed the training

C = Kevin receives certification

rule:

$T \rightarrow C$

Fact: T

Goal: C

→ CNF

$T \rightarrow C$

$\neg T \vee C$

Fact : T

Negated Goal : $\neg C$

→ Resolution

$\neg T \vee C$

Resolve with : T

Therefore : C

Resolve : C

with $\neg C$

Keith ~~receives a~~ certification

5. Access Control System

→ propositional logic

P = User enters the correct password

A = User is authenticated.

G = User is granted access

Rules :

$P \rightarrow A$

$A \rightarrow G$

Fact :

P

Goal :

G

CNF

Rule 1:

$P \rightarrow A$

$\neg P \vee A$

rule 2:

$A \rightarrow G$

$\neg A \vee G$

Fact:

P

Negated goal: $\neg G$

→ Resolution Step

Step 1:

$\neg P \vee A$

with P

gives : A

Step 2:

$\neg A \vee G$ with A

gives G

Step 3:

G with $\neg G$

gives $\{\}$

Since the empty clauses is obtained

The user is granted access.

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	20
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	20
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	23
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	19
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	14

96

100